FINAL PROJECT REPORT

Team ID:

Date: March 7, 2026

1. PROJECT TITLE

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau and Flask Integration.

2. PROJECT DESCRIPTION

- **System Overview:** This initiative provides a sophisticated end-to-end framework for retail intelligence. It synchronizes multidimensional sales data with interactive web technologies to empower stakeholders in making informed decisions regarding shop floor layouts and inventory positioning.
- **Problem Domain:** Merchandising Optimization and Spatial Retail Analytics.
- **Core Technology Used:** Tableau Desktop/Public, Python (Flask), and modern Frontend languages (HTML5/CSS3).
- **Key Capabilities:** High-fidelity dashboards, cross-functional filtering (Traffic/Promotions), and modular web embedding.
- **Target Users:** Retail Operations Leads, Category Strategists, and Supply Chain Analysts.
- **Real-world Relevance:** Replaces manual floor mapping with automated visual evidence, showing how specific shelf locations directly influence revenue.

3. APPLICATION SCENARIOS / USE CASES

- **Managerial Planning:** Store managers can evaluate seasonal performance by comparing the ROI of "Front-of-Store" displays against standard aisle locations.
- **Competitive Analysis:** Merchandising analysts can use scatter plots to identify pricing gaps relative to competitors and adjust placement strategy accordingly.

4. TECHNICAL ARCHITECTURE OVERVIEW

- **Data Layer:** Raw metrics are maintained within the `Product Positioning.csv` file.
- **Analytics Engine:** Tableau serves as the processing layer, handling calculated fields and visual rendering.
- **Cloud Layer:** Visualizations are hosted via Tableau Public to generate secure, interactive web embeds.
- **Backend Layer:** A Python Flask server manages the application logic and routing.

- **Frontend Layer:** A Bootstrap-based UI integrates the Tableau JavaScript API for a seamless user experience.

5. PREREQUISITES

- **Development Environment:** Tableau Desktop 2023+, VS Code or similar Python IDE.
- **Runtime Requirements:** Python 3.8+ and a standard modern web browser.
- **Dependencies:** Flask framework and Tableau JavaScript API libraries.

6. PRIOR KNOWLEDGE REQUIRED

- **Data Literacy:** Familiarity with retail KPIs such as Sales Volume and Promotion Lift.
- **Visualization Skills:** Basic proficiency in Tableau workbook and story creation.
- **Web Basics:** Fundamental understanding of Python routing and HTML iframe structures.

7. PROJECT OBJECTIVES

- **Technical:** Extract and sanitize insights from complex CSV datasets.
- **Performance:** Optimize the dashboard for sub-second filter response times.
- **Deployment:** Create a professional Flask wrapper for cloud-hosted Tableau content.

8. SYSTEM WORKFLOW

1. Initialize the local Flask server at `http://127.0.0.1:5000`.
2. The application routes the request to the `index.html` template.
3. The browser renders a unified UI with a custom navigation sidebar.
4. JavaScript functions (`showSection`) toggle between the Dashboard and Data Story views.
5. User interaction with filters triggers dynamic updates across all embedded charts.

9. MILESTONE 1: REQUIREMENT ANALYSIS

We focused on identifying the core variables within the `Product Positioning.csv`, such as Foot Traffic and Sales Volume. We then established a stable data connection in Tableau, ensuring proper categorical and geographic roles for each field.

10. MILESTONE 2: SYSTEM DESIGN

The project utilizes a responsive grid layout. This includes a KPI banner for high-level metrics and a centralized workspace for deep-dive visualizations, wrapped in a professional Bootstrap container.

11. MILESTONE 3: CORE IMPLEMENTATION

- **Data Prep:** We standardized currency formats and developed calculated fields like `[Price Variance]` to measure market competitiveness.
- **Visuals:** We created Bar Charts for volume analysis, Scatter Plots for pricing trends, and Heatmaps for demographic profiling.
- **Storytelling:** A 3-scene Tableau Story was built to guide users from a high-level overview to specific demographic deep-dives.

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

The `app.py` file was configured to handle site navigation. The frontend was styled with custom CSS variables to match a "Skill Wallet" theme, and interactive iframes were embedded to serve the Tableau Public content dynamically.

13. MILESTONE 5: TESTING & VALIDATION

We performed rigorous testing on filter latency and calculation accuracy. Test cases confirmed that choosing "High Traffic" zones correctly filtered all associated graphs and that navigation between scenes in the Story mode was smooth.

14. DEPLOYMENT

The architecture utilizes a hybrid approach: the visualizations are hosted on Tableau Public (Cloud), while the control interface runs on a local Flask server. Deployment involves publishing the workbook and updating the Flask templates with the new embed URLs.

15. PROJECT STRUCTURE

```
strategic_placement_project/
|-- data/ (Product Positioning.csv)
|-- tableau/ (Tableau1.twb)
|-- templates/ (index.html)
|-- app.py (Flask routing)
|-- requirements.txt (Python libs)
```

16. RESULTS

The final system successfully provides a centralized portal where retail managers can interact with complex positioning data without needing specialized analytical software.

17. ADVANTAGES & LIMITATIONS

- **Advantages:** Intuitive no-code interface, centralized data access, and highly responsive visuals.
- **Limitations:** Currently utilizes static data extracts; requires manual republication for database changes.

18. FUTURE ENHANCEMENTS

- **Live Data:** Integrating Snowflake or SQL Server for real-time data streaming.
- **Security:** Adding Flask-Login modules for secure, role-based user authentication.

19. CONCLUSION

This project proves that integrating Tableau's visual power with Flask's web flexibility creates a potent tool for retail optimization. It successfully transforms raw data into a strategic asset for business planning.

20. APPENDIX

- **Source Code:** Included below (`app.py` and `index.html`).
- **Documentation:** Detailed procedural steps are located in the project `README.md`.

SOURCE CODE

Flask Backend (app.py)

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

if __name__ == '__main__':
    app.run(debug=True, port=5000)
```

Frontend UI (index.html)

(Note: Use the standard Bootstrap and Tableau embedding code as previously defined in the documentation files.)

```
        Source Code: Frontend UI (index.html)
10.     <!DOCTYPE html>
11.     <html lang="en">
12.
13.     <head>
14.         <meta charset="UTF-8">
15.         <meta name="viewport" content="width=device-width, initial-scale=1.0"
16.         <title>Strategic Product Placement | Skill Wallet</title>
17.
18.         <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;4
19.         <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/boo
20.         <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-i
21.
```

```
        <style>
            :root {
                --primary-color: #0d6efd;
                --sidebar-bg: #111827;
                --sidebar-hover: #1f2937;
                --bg-color: #f3f4f6;
                --text-main: #1f2937;
                --text-muted: #6b7280;
            }

            body {
                font-family: 'Inter', sans-serif;
                background-color: var(--bg-color);
                overflow-x: hidden;
                color: var(--text-main);
            }

            /* Sidebar */
            .sidebar {
                height: 100vh;
                background-color: var(--sidebar-bg);
                color: white;
                padding-top: 24px;
                position: fixed;
                width: 260px;
                z-index: 100;
            }

            .brand-logo {
                display: flex;
                align-items: center;
                padding: 0 24px;
                margin-bottom: 2rem;
            }

            .brand-logo i {
                font-size: 1.8rem;
                color: #3b82f6;
                margin-right: 12px;
            }

            .brand-logo div {
                line-height: 1.2;
            }

            .brand-title {
                font-weight: 700;
                font-size: 1.1rem;
                letter-spacing: 0.5px;
            }

            .brand-subtitle {
                font-size: 0.75rem;
                color: #9ca3af;
                text-transform: uppercase;
                letter-spacing: 1px;
            }

```

```
80.             .nav-label {
81.                 font-size: 0.75rem;
82.                 color: #6b7280;
83.                 text-transform: uppercase;
84.                 letter-spacing: 1px;
85.                 margin: 1.5rem 24px 0.5rem;
86.                 font-weight: 600;
87.             }
88.
89.             .sidebar a {
90.                 color: #d1d5db;
91.                 text-decoration: none;
92.                 padding: 12px 24px;
93.                 display: flex;
94.                 align-items: center;
95.                 font-size: 0.95rem;
96.                 font-weight: 500;
97.                 transition: all 0.2s ease;
98.                 cursor: pointer;
99.                 border-left: 3px solid transparent;
100.            }
101.
102.            .sidebar a i {
103.                font-size: 1.1rem;
104.                margin-right: 12px;
105.                color: #9ca3af;
106.            }
107.
108.            .sidebar a:hover {
109.                background-color: var(--sidebar-hover);
110.                color: white;
111.            }
112.
113.            .sidebar a:hover i {
114.                color: white;
115.            }
116.
117.            .sidebar a.active {
118.                background-color: rgba(59, 130, 246, 0.1);
119.                color: #3b82f6;
120.                border-left: 3px solid #3b82f6;
121.            }
122.
123.            .sidebar a.active i {
124.                color: #3b82f6;
125.            }
126.
127.            /* Main Content */
128.            .main-content {
129.                margin-left: 260px;
130.                display: flex;
131.                flex-direction: column;
132.                min-height: 100vh;
133.            }
134.
135.            /* Top Navbar */
136.            .top-navbar {
137.                background-color: white;
```

```
138.                height: 70px;
139.                display: flex;
140.                align-items: center;
141.                justify-content: space-between;
142.                padding: 0 40px;
143.                border-bottom: 1px solid #e5e7eb;
144.            }
145.
146.            .search-bar {
147.                background-color: #f3f4f6;
148.                border-radius: 8px;
149.                padding: 8px 16px;
150.                display: flex;
151.                align-items: center;
152.                width: 300px;
153.            }
154.
155.            .search-bar input {
156.                border: none;
157.                background: transparent;
158.                outline: none;
159.                margin-left: 8px;
160.                width: 100%;
161.                font-size: 0.9rem;
162.            }
163.
164.            .user-profile {
165.                display: flex;
166.                align-items: center;
167.                gap: 12px;
168.                font-weight: 500;
169.                font-size: 0.9rem;
170.            }
171.
172.            .avatar {
173.                width: 36px;
174.                height: 36px;
175.                background-color: #3b82f6;
176.                color: white;
177.                border-radius: 50%;
178.                display: flex;
179.                align-items: center;
180.                justify-content: center;
181.                font-weight: 600;
182.            }
183.
184.            /* Page Content */
185.            .page-content {
186.                padding: 30px 40px;
187.                flex-grow: 1;
188.            }
189.
190.            .badge-skillwallet {
191.                background-color: #e0e7ff;
192.                color: #4338ca;
193.                padding: 4px 10px;
194.                border-radius: 20px;
195.                font-size: 0.75rem;
```

```
                font-weight: 600;
                margin-bottom: 10px;
                display: inline-block;
            }

            h2#main-title {
                font-weight: 700;
                font-size: 1.75rem;
                margin-bottom: 4px;
                color: #111827;
            }

            /* KPI Cards */
            .kpi-card {
                background: white;
                border-radius: 12px;
                padding: 20px;
                border: 1px solid #e5e7eb;
                box-shadow: 0 1px 3px rgba(0,0,0,0.05);
                display: flex;
                align-items: center;
                gap: 16px;
                transition: transform 0.2s ease;
            }

            .kpi-card:hover {
                transform: translateY(-2px);
                box-shadow: 0 4px 6px rgba(0,0,0,0.05);
            }

            .kpi-icon {
                width: 48px;
                height: 48px;
                border-radius: 10px;
                display: flex;
                align-items: center;
                justify-content: center;
                font-size: 1.4rem;
            }

            .kpi-icon.blue { background: #eff6ff; color: #3b82f6; }
            .kpi-icon.green { background: #f0fdf4; color: #22c55e; }
            .kpi-icon.purple { background: #faf5ff; color: #a855f7; }
            .kpi-icon.orange { background: #fff7ed; color: #f97316; }

            .kpi-details p {
                margin: 0;
                font-size: 0.85rem;
                color: var(--text-muted);
                font-weight: 500;
            }

            .kpi-details h4 {
                margin: 0;
                font-weight: 700;
                font-size: 1.25rem;
                color: #111827;
            }
```

```
254.
255.            /* Dashboard Container */
256.            .dashboard-wrapper {
257.                background: white;
258.                border-radius: 12px;
259.                padding: 0;
260.                box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -1p
261.                border: 1px solid #e5e7eb;
262.                min-height: 850px;
263.                overflow: hidden;
264.                margin-top: 24px;
265.            }
266.
267.            .view-section {
268.                display: none;
269.                animation: fadeIn 0.3s ease-in-out;
270.            }
271.
272.            .view-section.active {
273.                display: block;
274.            }
275.
276.            @keyframes fadeIn {
277.                from { opacity: 0; transform: translateY(5px); }
278.                to { opacity: 1; transform: translateY(0); }
279.            }
280.
281.            .btn-primary {
282.                background-color: #3b82f6;
283.                border-color: #3b82f6;
284.                font-weight: 500;
285.                padding: 8px 20px;
286.                border-radius: 6px;
287.            }
288.
289.            .btn-primary:hover {
290.                background-color: #2563eb;
291.                border-color: #2563eb;
292.            }
293.
294.        </style>
295.    </head>
296.
297.    <body>
298.
299.        <div class="sidebar">
300.            <div class="brand-logo">
301.                <i class="bi bi-wallet2"></i>
302.                <div>
303.                    <div class="brand-title">Skill Wallet</div>
304.                    <div class="brand-subtitle">Mini Project</div>
305.                </div>
306.            </div>
307.
308.            <div class="nav-label">Analytics Menu</div>
309.            <a onclick="showSection('dashboard')" id="nav-dashboard" class="c
310.                <i class="bi bi-grid-1x2-fill"></i>
311.                Placement Dashboard
```

```
            </a>

            <a onclick="showSection('story')" id="nav-story">
                <i class="bi bi-file-earmark-bar-graph-fill"></i>
                Data Story
            </a>
        </div>

        <div class="main-content">

            <div class="top-navbar">
                <div class="search-bar">
                    <i class="bi bi-search text-muted"></i>
                    <input type="text" placeholder="Search insights...">
                </div>
                <div class="user-profile">
                    <i class="bi bi-bell text-muted me-3" style="font-size: 1
                    <div class="avatar">DA</div>
                    <span>Data Analyst</span>
                </div>
            </div>

            <div class="page-content">
                <div class="d-flex justify-content-between align-items-end mb
                    <div>
                        <span class="badge-skillwallet"><i class="bi bi-check
                        <h2 id="main-title">Strategic Product Placement Analy
                        <p class="text-muted mb-0">
                            Analyzing the impact of store positions on sales
                        </p>
                    </div>
                    <div>
                        <button class="btn btn-primary">
                            <i class="bi bi-download me-2"></i>Export Report
                        </button>
                    </div>
                </div>

                <div class="row g-4 mb-2">
                    <div class="col-md-3">
                        <div class="kpi-card">
                            <div class="kpi-icon blue"><i class="bi bi-box-se
                            <div class="kpi-details">
                                <p>Total Products</p>
                                <h4>1,000</h4>
                            </div>
                        </div>
                    </div>
                    <div class="col-md-3">
                        <div class="kpi-card">
                            <div class="kpi-icon green"><i class="bi bi-shop"
                            <div class="kpi-details">
                                <p>Highest Traffic Zone</p>
                                <h4>Front of Store</h4>
                            </div>
                        </div>
                    </div>
                    <div class="col-md-3">
```

```
                        <div class="kpi-card">
                            <div class="kpi-icon purple"><i class="bi bi-tags
                            <div class="kpi-details">
                                <p>Active Promotions</p>
                                <h4>48%</h4>
                            </div>
                        </div>
                    </div>
                    <div class="col-md-3">
                        <div class="kpi-card">
                            <div class="kpi-icon orange"><i class="bi bi-grap
                            <div class="kpi-details">
                                <p>Avg. Sales Volume</p>
                                <h4>1,984</h4>
                            </div>
                        </div>
                    </div>
                </div>

                <div class="dashboard-wrapper">
                    <div id="dashboard" class="view-section active">
                        <iframe
                            src="https://public.tableau.com/views/Book1_17727
                            width="100%"
                            height="850"
                            style="border:none;">
                        </iframe>
                    </div>

                    <div id="story" class="view-section">
                        <iframe
                            src="https://public.tableau.com/views/Book1_17727
                            width="100%"
                            height="850"
                            style="border:none;">
                        </iframe>
                    </div>
                </div>

            </div>
        </div>

        <script>
            function showSection(sectionId) {
                // Hide all sections
                document.querySelectorAll('.view-section').forEach(section =>
                    section.classList.remove('active');
                });

                // Show selected section
                document.getElementById(sectionId).classList.add('active');

                // Remove active class from all links
                document.querySelectorAll('.sidebar a').forEach(link => {
                    link.classList.remove('active');
                });

                // Add active class to clicked link
```

```
                document.getElementById('nav-' + sectionId).classList.add('ac

                // Update Header Title based on selection
                const title = document.getElementById('main-title');
                title.innerText =
                    sectionId === 'dashboard'
                        ? 'Strategic Product Placement Analysis'
                        : 'FMCG Strategic Data Story';
            }
        </script>

        <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo

```